



The Automated Lead Generator

A Complete Guide to Building Your Own Privacy-First AI Marketing System

mercel-vercel.vercel.app

From Zero to Automated Lead Generation – Without Cloud

Dependency * The following document assumes the user has nothing established in terms of social accounts, or wants designated accounts specified strictly for business. * If you currently have some of these requirements, the installations can be omitted and you may utilize your current account providers. —

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If you find all of this overwhelming, or dont have time, you can hire us to do it, and we can set all of this up for your business. Feel free to contact us, including details of your current setup, and what your goals are, and we will be in contact with you with a quote.

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Chapter 1: Introduction

The Problem & The Solution

The Problem

Small businesses face a fundamental challenge: how to generate qualified leads without spending thousands on advertising or hiring a full-time marketing team. Cloud-based AI services offer a solution, but they come with serious drawbacks:

- **Data Privacy Risks:** Your customer data leaves your servers and goes to third parties.
- **Recurring Costs:** Subscription fees add up every month.
- **Generic Output:** Cloud AI is trained on everyone's data, so it produces generic content. (Everyones data becomes the property of the provider)
- **No Ownership:** You don't control the system or the data it processes.

The Solution

We built a solution that solves all these problems: a **privacy-first, automated lead generation system** that runs entirely on your own hardware. This guide is the complete documentation of what we did, step by step.

How the system operates:

- Uses **free, open-source software** (Ollama, OpenClaw, Codex).
- Runs on **your own hardware** (a laptop, a cloud server, or a spare PC).
- Costs **\$0 per month** in software subscriptions.
- Generates **content automatically** for your social media.
- Captures **leads** via your website.
- Sends **automated follow-up emails**.
- Requires **minimal maintenance**.

This isn't a theoretical guide. This is the exact system we built for our own business. We documented every step so you can do the same.

Chapter 2: The Big Picture

How It All Works

Here's what each component does:

Component	Purpose
OpenClaw	The orchestrator. It handles automation, scheduling, and connecting AI to your business tools.
Codex	A code-generation AI that helps you create content, scripts, and templates.
Ollama	The local AI model server. It runs LLMs like Llama 3.1, Qwen, etc., on your hardware.
Zernio	A bridge that connects OpenClaw to social media APIs (TikTok, Instagram, X).
Resend	Email delivery service (free tier). Sends emails to leads and notifications to you.
Upstash Redis	Token storage for your sample delivery system.
Tailscale	Secure remote access to your server from anywhere.
Firebase	User authentication and data storage (optional but useful).

This system is modular. You can use all of it, or just the parts you need. But for a complete lead generation system, you'll want all of these components working together.

Chapter 3: Hardware Requirements

What You Need

Before you start, you need to decide where your system will run. There are three main options:

Option	Hardware	Cost	Best For
Your Laptop	16-32 GB RAM, decent CPU	\$0 (you own it)	Testing and small-scale use
Cloud Server	Oracle Cloud Free Tier (4 ARM cores, 24 GB RAM)	\$0	24/7 operation, remote management
Dedicated Home Server	Used Dell PowerEdge, 64+ GB RAM	\$100-\$300 (one-time)	Heavy use, full control

Our Recommendation

For most small businesses, the **Oracle Cloud Free Tier** is the best option (If open spots are available). It's free, runs 24/7, and you can access it from anywhere. Here's what you get:

- **4 ARM cores** (Ampere A1)
- **24 GB RAM**
- **200 GB storage**
- **10 TB outbound data transfer**

This is more than enough to run Ollama, OpenClaw, and the rest of the stack.

If you can't get Oracle Cloud (their free tier: shape of 2 OCPUs and 12 GB of memory, is hard to sign up for), you can use: - A spare laptop (with Ubuntu Server installed) - A Raspberry Pi 5 (with 8 GB RAM) for lighter tasks - Any cloud provider with a free tier.

The key is **RAM**. Ollama models need RAM to run. An 8B parameter model needs about 8 GB. A 30B model needs 24+ GB. For a small business, 16-24 GB RAM is a good target.

Chapter 4: Software Stack

The Core Tools

Here's the complete list of software you'll use. Each tool has a specific role in the ecosystem.

4.1 OpenClaw (The Orchestrator)

Purpose: OpenClaw is the “brain” of your automation system. It connects your AI to your business tools and handles tasks like:

- Scheduling content posts
- Sending emails
- Generating proposals
- Managing social media
- Answering customer inquiries

It runs as a web server on your machine, and you interact with it via a command-line interface or a chat interface (like Telegram).

Why it matters: Without OpenClaw, you'd have to manually copy-paste between all your tools. OpenClaw connects everything.

4.2 Ollama (The AI Engine)

Purpose: Ollama is the software that runs your local AI models. It's a simple command-line tool that downloads and runs LLMs like:

- Llama 3.1 (8B, 70B)
- Qwen (7B, 14B, 30B)
- Mistral (7B, 22B)
- Any model from the Ollama library

Why it matters: Ollama gives you a private AI that runs on your hardware. No cloud dependency.

4.3 Codex (The Code Generator)

Purpose: Codex is an AI-powered code generator. It helps you create scripts, templates, and even full applications without writing code manually.

Why it matters: If you need a custom script or template, Codex can generate it for you. It works with Ollama as the model backend.

4.4 Zernio (The Social Media Bridge)

Purpose: Zernio connects OpenClaw to social media platforms. It handles the complex OAuth flows and rate limits for TikTok, Instagram, X, etc.

Why it matters: Without Zernio, OpenClaw couldn't post to social media. It's the bridge between your AI and your audience.

4.5 Resend (The Email Sender)

Purpose: Resend is a free email delivery service. It sends emails to your leads (download links, confirmations) and notifications to you (new lead alerts).

Why it matters: You can't send automated emails without an email service. Resend's free tier (3,000 emails/month) is perfect for small businesses.

4.6 Tailscale (The Secure Connection)

Purpose: Tailscale creates a secure VPN between your devices. You can SSH into your server from anywhere.

Why it matters: You need to manage your server remotely from a phone or laptop. Tailscale makes this secure and easy.

4.7 MAi-RAG-PA Knowledgebase (Acronym for: Memory Augmented Intelligence - Retrieval Augmented Generation - Personal Assistant)

Purpose: MAi-RAG-PA (which we developed) is a framework for building Retrieval-Augmented Generation (RAG) systems. It connects your AI to your documents (PDFs, emails, etc.) so it can answer questions about your business.

Why it matters: This turns a generic AI into a "digital Personal Assistant" that knows your business.

Chapter 5: Account Setup

Building the Ecosystem

Before you can set up the software, you need to create all the accounts that make up your ecosystem. Each account has a specific role.

5.1 Email Address (ProtonMail)

Purpose: This is your business email. You'll use it for: - Receiving lead notifications - Sending emails to clients - Registering for other accounts

Setup: 1. Go to proton.me 2. Create a free account (e.g., `yourbusiness@proton.me`) 3. Verify your email

Why ProtonMail? It's free, privacy-focused, and doesn't require a phone number for verification.

5.2 GitHub Account

Purpose: GitHub is where you'll store your code. It's essential for deploying to Vercel and managing your website.

Setup: 1. Go to github.com 2. Create a free account 3. Create a repository for your website (e.g., `my-website`)

5.3 Vercel Account

Purpose: Vercel is the hosting platform for your website. It connects to your GitHub repo and deploys automatically.

Setup: 1. Go to vercel.com 2. Sign up with your GitHub account 3. Connect your repository 4. Set up your domain (e.g., `yourbusiness.vercel.app`)

Note: If you already have a domain and hosting, you will not need a GitHub, or Vercel account.

5.4 Tailscale Account

Purpose: Tailscale gives you secure remote access to your server.

Setup: 1. Go to tailscale.com 2. Create a free account 3. Install Tailscale on your server and your workstation 4. Authenticate both devices

5.5 Zernio Account

Purpose: Zernio connects OpenClaw to social media platforms.

Setup: 1. Go to zernio.com 2. Create an account (free tier allows 2 platforms) 3. Connect your TikTok, Instagram, etc.

5.6 Resend Account

Purpose: Resend sends automated emails.

Setup: 1. Go to resend.com 2. Create a free account 3. Get your API key 4. Verify your domain (optional but recommended)

5.7 Formspree Account (Optional)

Purpose: Formspree handles form submissions on your website.

Setup: 1. Go to formspree.io 2. Create a free account 3. Create a form 4. Get your form endpoint URL

5.8 Social Media Accounts

Purpose: These are your content channels.

- **TikTok:** For short-form video content
- **Instagram:** For visual content and community building
- **X (Twitter):** For text-based updates and engagement
- **Telegram:** For direct communication and automation control

Setup: 1. Create accounts using your business email 2. Fill out bios with a clear value proposition 3. Add your website link to your bio

5.9 Upstash Redis (Optional)

Purpose: Stores tokens for your sample delivery system.

Setup: 1. Go to upstash.com 2. Create a free account 3. Create a Redis database 4. Copy the REST URL and Token

Chapter 6: Server Setup

Installing Ubuntu Server

Now it's time to set up your server. We will show you how to do this on Oracle Cloud (the free option), but the latter steps are similar for any Ubuntu Server.

6.1 Create an Oracle Cloud Account

1. Go to oracle.com/cloud/free
2. Sign up with your business email
3. Verify your identity (credit card for verification, no charge)
4. Wait for account approval (can take a few days)

6.2 Create a Compute Instance

1. Go to **Compute** → **Instances** → **Create Instance**
2. Choose an image: **Ubuntu 22.04** or **24.04**
3. Choose a shape: **VM.Standard.A1.Flex** (ARM, 4 OCPUs, 24 GB RAM)
4. Set boot volume to **50 GB** (free)
5. Add your SSH public key
6. Click **Create**

Note: Oracle Cloud's free tier ARM instances are often out of capacity. You may need to retry multiple times or upgrade to Pay As You Go (still free within limits).

6.3 Connect to Your Server

1. Download your SSH private key (created during instance setup)
2. Open a terminal on your workstation
3. Run:

```
ssh -i ~/.ssh/your-key ubuntu@<server-ip>
```

4. If you're using Tailscale, use the Tailscale IP for more secure access.

6.4 Configure the Server

Run these commands to update and prepare the server:

```
sudo apt update && sudo apt upgrade -y
sudo apt install build-essential git curl wget unzip -y
```

6.5 Install Tailscale (For Secure Remote Access)

```
curl -fsSL https://tailscale.com/install.sh | sh
sudo tailscale up --ssh --hostname=<your-server-name>
```

Note: Save the IP address that appears. This is your secure connection address.

Chapter 7: Software Installation

Step-by-Step

Now we'll install all the software on your server.

7.1 Install Ollama

Ollama is the AI model runtime. Install it with:

```
curl -fsSL https://ollama.com/install.sh | sh
```

Pull your first model:

```
ollama pull llama3.1:8b
```

This downloads a free, open-source 8B parameter model. It's a good starting point for most tasks.

7.2 Install OpenClaw

OpenClaw is the orchestrator. Install it with:

```
curl -fsSL https://openclaw.ai/install.sh | bash
```

This installs the command-line interface.

Configure the gateway:

```
openclaw config set gateway.bind loopback
openclaw config set gateway.auth.mode token
openclaw doctor --generate-gateway-token
```

Create a systemd service (so it runs automatically):

```

mkdir -p ~/.config/systemd/user
cat > ~/.config/systemd/user/openclaw.service <<'EOF'
[Unit]
Description=OpenClaw Gateway
After=network.target ollama.service

[Service]
Type=simple
ExecStart=/home/<user>/.openclaw/bin/openclaw gateway serve
Restart=on-failure
RestartSec=5
Environment="OLLAMA_API_KEY=ollama-local"

[Install]
WantedBy=default.target
EOF

systemctl --user daemon-reload
systemctl --user enable openclaw.service
systemctl --user start openclaw.service
sudo loginctl enable-linger <user>

```

7.3 Install Codex (Optional)

Codex helps generate code and templates. Since you may not have a ChatGPT subscription, you can configure it to use your local Ollama model.

1. Download the Codex binary from GitHub releases.
2. Move it to `/usr/local/bin/`.
3. Create a config file:

```

mkdir -p ~/.codex
cat > ~/.codex/config.toml <<'EOF'
model_provider = "ollama"
model = "llama3.1:8b"
[model_providers.ollama]
name = "Ollama Local"
base_url = "http://127.0.0.1:11434/v1"
wire_api = "responses"
EOF

```

7.4 Install MAi-RAG-PA (Optional)

If you want to build your own knowledge base system, install MAi-RAG-PA (available FREE on GitHub, our compliments). <https://mai-rag-pa.github.io/MAi-RAG-PA/> > **Note:** Read Documentation to See how this can be of use, and to further implement its capabilities.

```
curl -fsSL https://github.com/MAi-RAG-PA/MAi-RAG-PA/raw/main/  
install.sh | bash
```

If accessing remotely you will need authentication via ssh: ssh hostname@ip-address:8000 (example: ssh openclaw-server@192.168.1.17:8000) A desktop icon should have been created during installation. Clicking that will open your default browser for the interface. - *You May need to hard refresh the browser (Ctrl + Shift + R)

This gives you a complete RAG pipeline for your business documents with a Clean User Friendly Interface.

Chapter 8: Configuring the AI Brain

Now that the software is installed, you need to configure it to work together.

8.1 Connect OpenClaw to Ollama

OpenClaw needs to know where Ollama is running. Set the environment variable:

```
echo 'export OLLAMA_API_KEY="ollama-local"' >> ~/.bashrc  
source ~/.bashrc
```

Then restart OpenClaw:

```
systemctl --user restart openclaw.service
```

8.2 Choose Your Model

You have several models available in Ollama. For most business tasks:

- **llama3.1:8b** – Good general purpose model
- **qwen2.5:7b** – Fast, efficient
- **phi3:3.8b** – Lightweight, good for quick tasks
- **granite4.1:3b** – Excellent for tool calling

Set your default model in OpenClaw:

```
openclaw models set ollama/llama3.1:8b
```

8.3 Test Your System

Run a quick test to make sure everything is working:

```
openclaw agent --message "Hello, what model are you using?" --  
agent main --local
```

If it responds with your model name, you're good to go.

Chapter 9: Connecting Your Social Media

This is where the magic happens. You'll connect your social media accounts to OpenClaw via Zernio.

9.1 Set Up Zernio

1. Go to zernio.com
2. Create an account
3. Connect your TikTok, Instagram, etc.
4. Get your API key

9.2 Install Zernio Skill on OpenClaw

```
npx clawhub@latest install mikipalet/zernio-api
```

9.3 Configure Zernio in OpenClaw

```
openclaw config set plugins.entries.zernio.apiKey  
YOUR_ZERNIO_API_KEY
```

9.4 Test a Post

```
openclaw "Post a TikTok video about our new product"
```

OpenClaw will generate the content and post it via Zernio.

Chapter 10: Setting Up Lead Capture

This is how you collect emails from potential customers.

10.1 Set Up Your Website (Vercel)

1. Create a Next.js website (We documented this in detail earlier).
2. Deploy to Vercel.
3. Add a “Free Sample” page with an email capture form.

10.2 Configure Email Sending (Resend)

1. Get your Resend API key.
2. Add it to your Vercel environment variables.
3. Create an API route to send emails (e.g., `app/api/send-sample/route.ts`).

10.3 Store Lead Data (Upstash Redis)

1. Create an Upstash Redis database.
2. Add the REST URL and Token to Vercel environment variables.
3. Use Redis to store tokens for your download links.

10.4 The Complete Flow

1. User visits your website.
 2. They submit their email to get a free sample.
 3. Your server generates a unique, time-limited download link.
 4. Resend sends the link to the user’s email.
 5. You receive a notification in your inbox.
 6. The user clicks the link and downloads the file.
 7. The link expires after 24 hours (or one use).
-

Chapter 11: Automating Your Content

This is where you set up automatic content generation and posting.

11.1 Create a Content Calendar

OpenClaw can generate content on a schedule. Use the cron feature:

```
openclaw cron add "daily-post" --schedule "0 9 * * *" --task  
"Generate a TikTok caption about AI for small businesses  
and post via Zernio."
```

11.2 Create Content Templates

To get consistent quality, create templates for common content types:

- **Sales proposals:** “I’ve attached a proposal for [project]. Here’s what we’ll deliver and the timeline.”
- **Follow-up emails:** “Hi [name], just checking in about [topic]. Any questions?”
- **Social media posts:** “Did you know [fact]? This is why your business needs [solution].”

OpenClaw can fill in these templates with your business information.

11.3 Generate Content with Local AI

Use Ollama to generate content. For example:

```
ollama run llama3.1:8b  
"Write a 100-word product description for a construction  
company that specializes in residential renovations."
```

11.4 Post Automatically

Once you’ve configured Zernio, OpenClaw can post to social media automatically. Set up a schedule so you don’t have to think about it.

Chapter 12: Maintenance & Daily Operation

This is how you run the system on a day-to-day basis.

12.1 Daily Tasks

- **Check your email** (leads, notifications)
- **Review content** that OpenClaw generated (make sure it’s appropriate)
- **Engage with comments** on your social media posts (manually or via OpenClaw)

12.2 Weekly Tasks

- **Update your content calendar** (if needed)
- **Review analytics** (which posts performed best?)
- **Adjust your models** (if the AI is generating poor content)
- **Backup your system** (tokens, config files)

12.3 Monthly Tasks

- **Update your software** (Ollama, OpenClaw, etc.)
- **Review your lead database** (which leads converted? Which didn't?)
- **Optimize your funnel** (change your free sample, adjust your pricing, etc.)

12.4 Monitoring Your System

Use `systemctl status` to check if services are running:

```
systemctl --user status openclaw.service  
systemctl --user status ollama.service
```

If something stops, the system should restart automatically (via `Restart=on-failure`). But you should check regularly.

12.5 Interacting with OpenClaw

You can interact with OpenClaw via: - **Telegram**: Set up a bot and chat with it directly. - **CLI**: Run commands in a terminal. - **Web UI**: Access the control panel via a browser (if configured).

Example Telegram interaction: - “Post a TikTok about our new service” - “Draft a proposal for [client name]” - “Summarize this document” (attach PDF) - “Check our leads for today”

Chapter 13: Troubleshooting Common Issues

Here are the most common problems and how to fix them.

13.1 OpenClaw Won't Start

Symptoms: Service fails to start, or you see errors.

Fixes: 1. Check if Ollama is running: `systemctl --user status ollama.service` 2. Check if the port is in use: `ss -tlnp | grep 3000` 3. Ensure the `OLLAMA_API_KEY` is set: `systemctl --user show openclaw.service | grep Environment` 4. Restart the service: `systemctl --user restart openclaw.service`

13.2 AI Model Not Responding

Symptoms: OpenClaw says “deterministic mode” or won’t use your model.

Fixes: 1. Verify Ollama is running: `curl http://127.0.0.1:11434/api/tags` 2. Ensure the model is pulled: `ollama list` 3. Set the correct model: `openclaw models set ollama/llama3.1:8b` 4. Restart OpenClaw: `systemctl --user restart openclaw.service`

13.3 Social Media Posting Fails

Symptoms: OpenClaw can’t post to TikTok/Instagram.

Fixes: 1. Check your Zernio connection: `openclaw status` 2. Re-authenticate your social accounts in Zernio. 3. Ensure your API key is correct: `openclaw config get plugins.entries.zernio.apiKey` 4. Test a simple post: `openclaw "Post 'Hello' to TikTok"`

13.4 Emails Not Sending

Symptoms: Leads don’t receive their download link.

Fixes: 1. Check your Resend API key in Vercel. 2. Verify your domain is verified in Resend (for sending to real users). 3. Check Resend logs for errors. 4. Test with a simple email: `curl -X POST https://api.resend.com/emails -H "Authorization: Bearer <key>" -d '{"from": "test@example.com", "to": "test@example.com", "subject": "Test", "html": "<p>Hello</p>"}'`

13.5 Server Running Low on Memory

Symptoms: Ollama crashes, or system becomes slow.

Fixes: 1. Reduce the model size (e.g., use a 3B model instead of 8B). 2. Increase the swap space. 3. Stop unused services. 4. Restart the server (if persistent).

Chapter 14: Scaling Your System

Once you have the basics working, here's how to grow.

14.1 Add More Social Platforms

Zernio's free tier allows 2 platforms. As you grow, you can: - Connect more platforms (X, Reddit, LinkedIn). - Upgrade Zernio for unlimited connections.

14.2 Add More Content Types

Start with text posts. Add: - Image posts (generate graphics with AI). - Video posts (use AI video generators). - Podcasts (use TTS to convert text to audio).

14.3 Build a Knowledge Base

Use MAi-RAG-PA to store your business documents: - FAQs - Product catalog - Client testimonials - Case studies

This makes your AI smarter about your business.

14.4 Automate Email Sequences

Set up automated email sequences for leads: - Welcome email (instant) - Follow-up email (day 2) - Product offer (day 4) - Special discount (day 7)

These can all be handled by OpenClaw + Resend.

14.5 Add Payment Processing

Connect Stripe or PayPal to your website to accept payments. This closes the loop from lead → customer.

Chapter 15: The Complete Checklist

Here's the complete, step-by-step checklist to build this system.

Phase 1: Preparation

- ☐ Create ProtonMail email
- ☐ Create GitHub account
- ☐ Create Vercel account
- ☐ Create Tailscale account
- ☐ Create Zernio account
- ☐ Create Resend account
- ☐ Create social media accounts (TikTok, Instagram, X, Telegram)
- ☐ Set up your business domain (optional)

Phase 2: Server Setup

- ☐ Choose your hardware (cloud server, laptop, etc.)
- ☐ Install Ubuntu Server
- ☐ Configure SSH access
- ☐ Install Tailscale for remote access

Phase 3: Software Installation

- ☐ Install Ollama
- ☐ Pull a model (llama3.1:8b or similar)
- ☐ Install OpenClaw
- ☐ Install Codex (optional)
- ☐ Install MAi-RAG-PA (optional)
- ☐

Set up systemd services

Phase 4: Configuration

☐

Connect OpenClaw to Ollama

☐

Set default model

☐

Test AI responses

☐

Set up Zernio connection

☐

Configure social media posting

Phase 5: Website Setup

☐

Create Next.js website

☐

Deploy to Vercel

☐

Add sample page with email capture

☐

Set up Resend email sending

☐

Set up Upstash Redis (for tokens)

Phase 6: Automation

☐

Create content templates

☐

Set up cron jobs for posting

☐

Configure email sequences

☐

Test the complete flow

Phase 7: Launch

☐

Post your first content

☐

Collect your first lead

☐

Send your first follow-up email

☐

Make your first sale

Chapter 16: Conclusion

You've now built a complete, privacy-first, automated lead generation system!

The key takeaways:

1. **You own everything** – Your data stays on your hardware.
2. **It's free to run** – No monthly subscriptions for software.
3. **It's automated** – You don't have to manually post or follow up.
4. **It's adaptable** – You can add more features as you grow.
5. **You can do this** – You don't need to be a programmer.

This system took us a week to build, but we have documented every step. Now you can do it in a few days.

The hardest part isn't the setup. It's learning to trust the system. Trust that OpenClaw will post good content. Trust that Ollama will write good proposals. Trust that your leads will convert.

But once you trust it, you'll have a 24/7 marketing machine that works while you sleep.

Your next step: Start building. Even if you only get through Phase 1 today, that's progress.

If you find all of this overwhelming, or dont have time, you can hire us to do it, and we can set all of this up for your business. Feel free to contact us including details of your current setup, and what your goals are, and we will be in contact with you with a quote.

****Contact us at: mercel-vercel.vercel.app to learn more.**

About Mercel

Mercel is dedicated to helping businesses leverage privacy-first AI automation. We build custom AI systems that run on your own hardware, ensuring your data never leaves your servers. We also provide comprehensive guides and consulting services for those who want to build their own systems.

Visit us at mercel-vercel.vercel.app to learn more.

This guide was formatted using a local AI assistant running on a personal server. No data was shared with third-party cloud services.